

# Role Stressors and Customer-Oriented Boundary-Spanning Behaviors in Service Organizations

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*The authors investigate three types of customer-oriented boundary-spanning behaviors (COBSBs) a frontline service employee may perform that are associated with linking a service organization to its potential or actual customers: external representation, internal influence, and service delivery. The authors propose and test a withdrawal model to explain the negative effects of role conflict and role ambiguity on COBSBs across a sample of 220 lower-level, nonprofessional service providers of a major retail bank and a sample of 90 higher-level, professional service providers from the business credit division of an international financial services corporation. The results demonstrate that (1) indirect paths through job satisfaction and organizational commitment entirely account for the negative effects of the role stressors on COBSBs, (2) the indirect negative effects of the role stressors are stronger on external representation and internal influence behaviors, and (3) role conflict also has a significant positive direct relationship with internal influence behaviors.*

**Keywords:** *customer-oriented boundary-spanning behaviors; role conflict; role ambiguity; service organization; organizational citizenship*

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It is widely acknowledged among both researchers and practitioners (cf. Rucci, Kirn, and Quinn 1998; Singh 2000) that service quality in many industries depends on

the customer-oriented behaviors of frontline service employees (FSEs). As Zeithaml, Berry, and Parasuraman (1988) observed, "Service quality is highly dependent on the performance of [frontline] employees" (p. 35). Similarly, Hartline, Maxham, and McKee (2000) argued that "in many cases, customer contact employees are the first and only representation of a service firm" (p. 35). Yet, despite the important role that FSEs play in linking the organization to its customers, few studies have considered antecedents of FSE customer-oriented behaviors, especially behaviors that extend beyond the interpersonal service encounter (cf. Bettencourt, Gwinner, and Meuter 2001).

The services marketing/management and boundary-spanning literatures both suggest three key dimensions of customer-linking behaviors of FSEs that contribute to the development of market-driven organizational capabilities and resources (cf. Day 1994). First, FSEs play an important part in representing the organization to outsiders (including customers) and enhancing the firm's image and legitimacy through their advocacy of the firm and its products and services (Adams 1976; Aldrich and Herker 1977; Booms and Bitner 1981; Bowen and Schneider 1985). Second, the boundary-spanning position of FSEs provides them many opportunities to share information internally about evolving customer needs and possible improvements in service delivery (Aldrich and Herker 1977; Schneider and Bowen 1984; Zeithaml et al., 1988). Third, service quality perceptions and customer satisfaction are largely dependent on frontline employee service delivery behaviors such as courtesy, personal attentiveness, responsiveness, and keeping promises (Aldrich and Herker 1977; Parasuraman, Zeithaml, and Berry 1988).

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Drawing from these literatures, we investigate three types of boundary-spanning behaviors the FSE may perform that are associated with linking the organization to its potential or actual customers (cf. Brief and Motowidlo 1986): (1) *external representation*, being vocal advocates to outsiders of the organization's image, products, and services (cf. Booms and Bitner 1981; Bowen and Schneider 1985); (2) *internal influence*, taking individual initiative in communications to the firm and coworkers to improve service delivery by the organization, coworkers, and oneself (cf. Schneider and Bowen 1984; Zeithaml et al. 1988); and (3) *service delivery*, serving customers in a conscientious, responsive, attentive, and courteous manner (cf. Parasuraman et al. 1988).

Included among our three dimensions are customer-oriented boundary-spanning behaviors (COBSBs) that are likely to be considered relatively more role prescribed and those that are likely to be considered relatively more extrarole. For the majority of FSEs, service delivery behaviors are likely to be relatively more role prescribed due to their frequent specification in job descriptions, training materials, and performance evaluation forms. External representation and internal influence behaviors, on the other hand, are likely to be relatively more discretionary (i.e., relatively more extrarole; cf. Organ 1988a). This assertion is reinforced in the two service contexts we study by the presence of service delivery behaviors in internal documents such as job descriptions, training materials, and performance evaluation forms. Still, we emphasize the *relative* degree to which each type of customer-oriented behavior is role prescribed versus extrarole because recent research suggests that it may be more appropriate to view a variety of pro-social organizational behaviors, including traditional citizenship behaviors, as existing on a discretion continuum from entirely role prescribed to entirely extrarole with the majority of behaviors lying somewhere between the two extremes (Tepper, Lockhart, and Hoobler 2001). Furthermore, the role-prescribed/extrarole distinction is likely to be especially fuzzy for service employees and companies due to the difficulty in precisely specifying FSE behaviors as they interface with customers (cf. Hartline et al. 2000; Singh 2000; Zeithaml et al. 1988).

The objective of our study is to test the applicability of a traditional theoretical model of the employee turnover process in explaining the detrimental impact of role stressors on COBSBs. Specifically, many studies in management and marketing hold that role conflict and role ambiguity have a negative impact on employee turnover through their indirect, negative effects on job satisfaction and/or organizational commitment (cf. Jones, Kankak, Futrell, and Johnston 1996; Netemeyer, Burton, and Johnston 1995). That is, the turnover process is proposed as one in which role stress leads to psychological withdrawal in the form of reduced job satisfaction and/or

organizational commitment, which in turn leads to behavioral withdrawal from the organization. Recent research by MacKenzie, Podsakoff, and Ahearne (1998) and Chen, Hui, and Sego (1998) has demonstrated that reduced performance of traditional organizational citizenship behaviors precedes actual employee turnover, suggesting that theoretical models used to understand employee turnover may also be useful for explaining behavioral withdrawal from the organization in the form of other relatively discretionary pro-social organizational behaviors such as COBSBs.<sup>1</sup> As such, we test a model in which role stressors are hypothesized to have an indirect, negative effect on COBSBs via job satisfaction and/or organizational commitment.

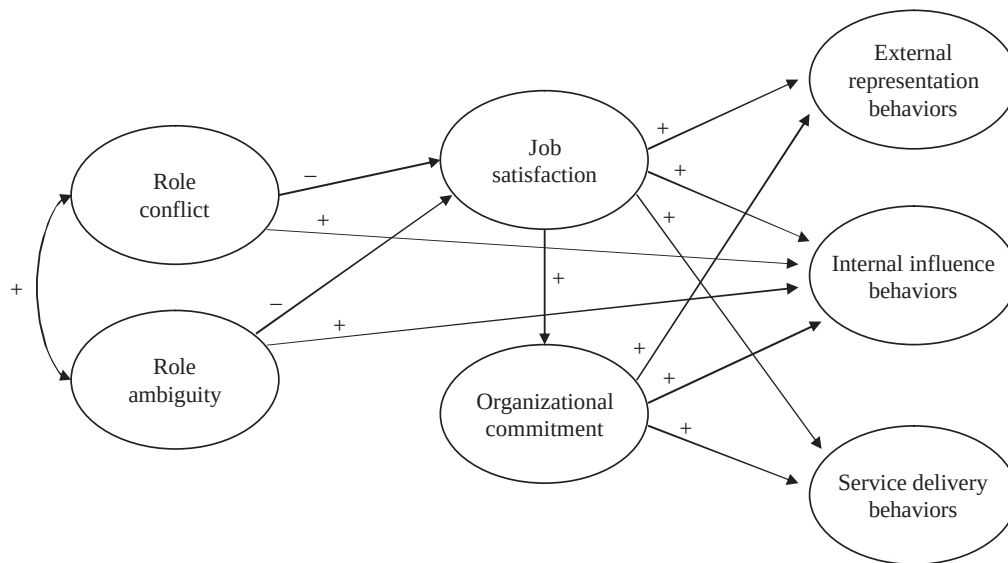
In testing the applicability of a withdrawal model for understanding COBSBs, our research addresses several important issues. First, it is not evident whether the behavioral withdrawal process will extend to explaining withdrawal in the form of COBSBs. Second, it is not clear if different dimensions of COBSBs might be more or less likely to be reduced in response to perceived role stress as mediated by job satisfaction and/or organizational commitment. Recent research by Tepper et al. (2001), however, suggests that more extrarole forms of COBSBs might be more strongly affected by the withdrawal process initiated by role conflict and role ambiguity. Third, although withdrawal models highlight the possible indirect, detrimental effects of role stress on COBSBs, some research that we review subsequently (cf. Kahn, Wolfe, Quinn, and Snoek 1964; Singh, Goolsby, and Rhoads 1994) also suggests the possibility of direct, positive effects of the role stressors on distinct forms of COBSBs—especially the more constructive, extrarole forms. Finally, because some past research indicates that characteristics of employee samples (e.g., organizational level, professional status, gender) may moderate relationships among role stressors and their consequences, we investigate our theoretical model simultaneously among two distinct samples of FSEs: (1) a retail banking sample consisting predominantly of low-level, nonprofessional, female service employees; and (2) a business credit sample consisting predominantly of high-level, professional, male service employees.

Following development of our hypotheses, the methodology and results of our empirical research are discussed. We conclude the article with a discussion of the findings and consideration of implications for managers and future research.

## MODEL AND HYPOTHESES

Figure 1 presents the theoretical model we investigate. We first develop hypotheses concerning the negative indirect effects of role conflict and role ambiguity on each of

**FIGURE 1**  
**A Model of Relationships Between Role Stressors, Job Satisfaction, Organizational Commitment, and Customer-Oriented Boundary-Spanning Behaviors**



the three COBSB dimensions through job satisfaction and/or organizational commitment. Next, we propose that the indirect, negative path relationships from role conflict and role ambiguity to external representation and internal influence are stronger than the indirect paths from the role stressors to service delivery behaviors. Finally, we hypothesize that role conflict and role ambiguity will have positive, direct relationships with internal-influence COBSBs when the negative, indirect relationships between these variables are controlled statistically.

## Hypotheses

Role conflict and role ambiguity are among the most commonly studied workplace variables that might lead to employee psychological or behavioral withdrawal from the job or organization (cf. Jackson and Schuler 1985). Role conflict refers to perceived incompatibility among role expectations and demands (e.g., from supervisors, customers, policies) that impinge on employee role performance (Rizzo, House, and Lirtzman 1970). Role ambiguity occurs when an employee lacks the information needed to effectively enact his or her role (Rizzo et al. 1970). As sources of workplace stress, role conflict and role ambiguity are proposed to be negatively related to psychological withdrawal in the form of reduced job satisfaction (the employee's overall evaluation of his or her job situation and experiences) and organizational commitment (psychological attachment felt by the employee for the firm) because employees desire clear and nonconflicting paths to performance and rewards (cf. Dubinsky and Hartley

1986a, 1986b). Meta-analyses in both marketing and management have provided considerable evidence that both role conflict and role ambiguity are significantly, negatively related to both job satisfaction and organizational commitment (Brown and Peterson 1993; Jackson and Schuler 1985; Mathieu and Zajac 1990), although some theory and empirical evidence suggests that job satisfaction fully mediates the relationship between the role stressors and organizational commitment, as illustrated in Figure 1 (cf. Hampton, Dubinsky, and Skinner 1986).

Consistent with Goolsby (1992), we contend that role stressors are linked to behavioral withdrawal indirectly through their effect on psychological withdrawal responses. Included among the psychological withdrawal responses in Goolsby's (1992) review of the literature are reduced job satisfaction and reduced organizational commitment. Indeed, theoretical and empirical research generally provide support for an indirect effect of role conflict and role ambiguity on behavioral withdrawal in the form of turnover via reduced job satisfaction and/or organizational commitment. Netemeyer et al. (1995), for example, empirically compared four models of role stressor consequences derived from the management literature and concluded that the best model was one in which role conflict and role ambiguity only influenced employee turnover indirectly via job satisfaction and organizational commitment.

Recent management research finds that employees may choose to withdraw from the job and/or organization by withholding certain functional behaviors while still maintaining organizational membership (cf. Chen et al. 1998;

MacKenzie et al. 1998), and these additional behavioral withdrawal responses are also related to job satisfaction and organizational commitment (cf. Organ and Ryan 1995). As Goolsby (1992) argued, behavioral withdrawal itself is a process, "only the final stage of which is turnover" (p. 160). In fact, the studies of Chen et al. (1998) and MacKenzie et al. (1998) suggest that *reduced* performance of various pro-social organizational behaviors are an early and partial form of behavioral withdrawal from the job or firm that precedes complete withdrawal in the form of turnover.

Although other possible mediators of the withdrawal effects of role stressors might have been considered, we focus only on job satisfaction and organizational commitment for several reasons. First, as discussed, job satisfaction and organizational commitment are among the most common psychological withdrawal mediators of the effects of role conflict and role ambiguity on employee turnover. Second, job satisfaction and organizational commitment are among the most widely studied social exchange attitudinal antecedents of various prosocial organizational behaviors in the management (cf. Organ and Ryan 1995) and marketing literatures (cf. MacKenzie et al. 1998; Netemeyer, Boles, McKee, and McMurrian 1997). In fact, Organ (1988b) has argued that some negative aspects of the employee's relationship with the organization (e.g., role conflict and role ambiguity) may prompt the employee to redefine the nature of his or her relationship with the firm from one based more on social exchange (characterized by high commitment, satisfaction, and open-ended role expectations) to one based more on contractual or economic exchange (characterized by low commitment, satisfaction, and limited voluntary contributions). Third, our research focuses on the mediating effects of *both* job satisfaction and organizational commitment, not only because there is reason to believe that each of them individually might mediate the negative effects of role stressors on COBSBs but also because they are known to be related to each other. Job satisfaction is generally believed to cause organizational commitment in part because it is more specific, less stable, and more rapidly formed, and the empirical evidence overwhelmingly supports this view (cf. Brown and Peterson 1993; Steers 1977; Williams and Hazer 1986). This suggests that the mediating effects of these two constructs will be intertwined. The prior discussion leads to the following hypotheses.

*Hypothesis 1a:* Role conflict will be negatively related to external representation due to its effects on job satisfaction and/or organizational commitment.

*Hypothesis 1b:* Role conflict will be negatively related to internal influence due to its effects on job satisfaction and/or organizational commitment.

*Hypothesis 1c:* Role conflict will be negatively related to service delivery due to its effects on job satisfaction and/or organizational commitment.

*Hypothesis 2a:* Role ambiguity will be negatively related to external representation due to its effects on job satisfaction and/or organizational commitment.

*Hypothesis 2b:* Role ambiguity will be negatively related to internal influence due to its effects on job satisfaction and/or organizational commitment.

*Hypothesis 2c:* Role ambiguity will be negatively related to service delivery due to its effects on job satisfaction and/or organizational commitment.

Although we expect the negative effects of role conflict and role ambiguity to be indirectly related to all three COBSB dimensions, we also hypothesize that the negative effects of the two role stressors on external representation and internal influence behaviors will be stronger than the negative effects of the two role stressors on service delivery behaviors, and that these negative differential relationships will also be mediated by job satisfaction and/or organizational commitment on the basis of similar withdrawal reasoning. Jex (1998), for example, contended that the negative effects of stress on employee behaviors may be most evident in reduced contribution of more extrarole behaviors, such as external representation and internal influence behaviors. More extrarole forms of employee behaviors offer the employee a behavioral withdrawal response in the presence of role conflict and role ambiguity that does not threaten their receipt of formal organizational rewards and allows them to maintain an outward appearance of commitment to the firm (cf. Griffeth, Gaertner, and Sager 1999; Organ 1988b).

Although the prior discussion provides a basis for expecting stronger negative relationships between role conflict and role ambiguity and external representation and internal influence behaviors, we also contend that these differential negative relationships work through a role stress → psychological withdrawal → behavioral withdrawal process. More than a decade ago, Organ (1988a, 1988b) contended that on the continuum from more to less discretionary organizational behaviors, it is the more discretionary behaviors that are likely to be withheld as a relationship becomes defined in more contractual exchange terms (i.e., lowered social exchange evaluations in terms of job satisfaction and/or organizational commitment). That is, more discretionary forms of COBSBs are more likely to be withheld as behavioral withdrawal responses in the face of reduced job satisfaction or organizational commitment because these forms of COBSBs are more likely to be viewed as social exchange gestures (Blau 1964; Organ 1988a, 1988b). This accounts for why a meta-analysis by Organ and Ryan (1995) revealed a stronger relationship between job satisfaction and citizenship behav-

iors than previous meta-analyses had identified between job satisfaction and traditional in-role measures of employee performance. On the basis of this discussion, we offer the following hypotheses:

*Hypothesis 3a:* The indirect negative effect of the role stressors on external representation (mediated by job satisfaction and/or organizational commitment) will be stronger than the indirect negative effect of the role stressors on service delivery.

*Hypothesis 3b:* The indirect negative effect of the role stressors on internal influence (mediated by job satisfaction and/or organizational commitment) will be stronger than the indirect negative effect of the role stressors on service delivery.

The traditional argument concerning the relationship between role stressors and job performance is that conflicting and unclear expectations inhibit performance because employee effort is compromised and misdirected (Rizzo et al. 1970). Inconsistent expectations (role conflict) and vague understanding of job requirements (role ambiguity) are considered antithetical to effective performance. However, multiple studies of boundary-spanning employees have identified either negative, positive, or nonsignificant relationships between both role conflict and role ambiguity and employee performance (cf. Babin and Boles 1996; Dubinsky and Hartley 1986a, 1986b; Hampton et al. 1986; Singh 1998), suggesting that both positive and negative role stress effects may be simultaneously operating.<sup>2</sup>

The theoretical basis for a possible positive relationship between role stress and employee performance, although often overlooked, has actually been recognized since the discussion of Kahn et al. (1964). As they concluded,

To regard conflict simply as a disruption of an otherwise harmonious way of life is to overlook the fact that conflict often provides the basis for individual achievement and social progress. The same can be said for ambiguity, for while ambiguity implies a disorderliness that is antithetical to the very idea of organization, it also permits a kind of flexibility that can facilitate adaptation to changing circumstances. (P. 54)

Consistent with this view, Weatherly and Tansik (1993) found that FSEs often responded to role stress by working harder to satisfy work demands and engaging in extrarole behaviors.

Although prior research suggests the possibility of competing negative and positive effects of role stressors on employee performance behaviors, the majority of past empirical studies have not been able to separate these competing effects because the role stressors were generally modeled *only* as direct antecedents of performance (cf.

Babin and Boles 1996; Dubinsky and Hartley 1986a, 1986b; Hampton et al. 1986; Singh 1998), meaning the models were essentially misspecified (see Singh et al. 1994 for a notable exception using a global, self-rated performance measure).<sup>3</sup> In contrast, our study models the negative indirect effects of role conflict and role ambiguity on COBSBs via job satisfaction and organizational commitment, allowing us to empirically distinguish any competing effects of the role stressors on COBSBs.<sup>4</sup> Past studies have also not considered how distinctions between different types of employee performance behaviors might affect whether positive effects are identified in addition to negative effects. In contrast, we propose that in addition to negative indirect effects on all three COBSB dimensions, the role stressors are likely to have positive direct effects on internal influence behaviors due to their more extrarole, change-oriented focus (cf. Van Dyne et al. 1994).

As discussed previously, the negative indirect paths from role conflict and role ambiguity to COBSBs reflect a withdrawal response of employees, as suggested by Goolsby (1992). While some employees may respond to role stressors with psychological and behavioral withdrawal from the job and organization, Goolsby (1992) also argued that some employees might respond to role stress more constructively by seeking “to control, alter, and manage the situation that is creating the stress” (Goolsby 1992, p. 158). Hirschman (1970) similarly considered exit (i.e., behavioral withdrawal) and voice (i.e., vocal attempts to change organizational conditions) distinct behavioral responses an employee might select when faced with objectionable workplace conditions. Thus, whereas some employees may respond to role stressors negatively via calculated reductions of customer-oriented behaviors, other employees may respond positively with citizenship-type behaviors designed to change the situation (Griffeth et al., 1999).

Thus, although we do not explicitly test for differences in responses across employee types, prior research provides the basis for hypothesizing positive direct effects of role conflict and role ambiguity on internal influence behaviors (i.e., vocal extrarole attempts at change) in addition to negative indirect effects. This discussion leads to our final set of hypotheses.

*Hypothesis 4a:* After accounting for indirect negative effects of role conflict on internal influence (mediated by job satisfaction and/or organizational commitment), role conflict will have a positive direct effect on internal influence.

*Hypothesis 4b:* After accounting for indirect negative effects of role ambiguity on internal influence (mediated by job satisfaction and/or organizational commitment), role ambiguity will have a positive direct effect on internal influence.

Although we do not offer hypotheses concerning direct positive effects from the role stressors to external representation and service delivery behaviors due to limited theoretical underpinning, we recognize this possibility in our analyses through tests of the overidentifying restrictions of Figure 1.

## METHOD

### Samples and Procedures

**Retail banking.** Our first sample consisted of boundary-spanning service employees of a major, multistate retail bank. We restricted our sample to two nonmanagement service positions that primarily consisted of full-time, salaried employees. Study materials were distributed in packets to branch managers, who then distributed individual materials (survey, postage-paid return envelope, and letter assuring confidentiality) to all employees of the sampled positions in their branches. The employee survey contained measures of role conflict, role ambiguity, job satisfaction, and organizational commitment, along with self-report measures of the three forms of COBSBs. Managers also completed a survey asking them to rate the COBSBs for each of the sampled employees in their branch. To match the surveys from individual FSEs and managers, employees were requested to write in their first and last name on their survey, and managers were asked to write the name of the individual employee being evaluated on each survey they completed.

Approximately 400 FSEs were surveyed across 106 branches, representing all stand-alone branches within two states in which the bank operates.<sup>5</sup> Completed surveys were received from 258 employees for a response rate of 64 percent. It was possible to match 220 completed employee surveys with a manager rating of COBSBs. Thus, the matched response rate was 55 percent. As expected, 77 percent of the sample consisted of full-time, salaried employees. Approximately 93 percent of the employee sample were female. The median age was 37. The majority of the sample (58%) reported "some college" as their highest level of education; only 15 percent of the sample had a college degree. The median organizational tenure was high among this sample at 99 months (more than 8 years). A check for nonresponse bias in which the date a survey was received by the researchers was regressed on each of the model variables (e.g., job satisfaction), and selected demographic variables (e.g., tenure, age) produced a nonsignificant overall model. Of the branch managers providing COBSB evaluations, 79 percent were female. The median age of managers was 43. The majority of managers (52%) reported "some college" as their highest level of education, while 26 percent had a

college degree. Managers' median organizational tenure was 18 years.

**Business credit.** Our second sample consisted of professional boundary-spanning service providers of the business credit division of an international financial services corporation. Given the professional nature of the sample, study materials (letter, survey, postage-paid return envelope) were sent directly to individual employees. Employee surveys were coded so that they could be matched with manager responses. As in the retail banking context, the employee survey contained measures of role conflict, role ambiguity, job satisfaction, and organizational commitment, along with self-report measures of the three forms of COBSBs. Managers also received an additional packet of surveys and return envelopes in order to be able to evaluate COBSBs of individual employees. As before, managers were asked to write the name of the individual service employee being evaluated on each survey they completed.

Given the small size of the firm (125 total employees), all 97 professional employees who had extensive client interaction were included in the sample; as such, the sample is quite diverse, consisting of loan originators (22%), underwriters (13%), auditors (5%), relationship managers (19%; a position responsible for managing the client relationship), loan administrators (15%), and managers of these positions (26%). All field managers were also included in the employee sample. The company president provided COBSB ratings on his direct reports.

We received completed surveys from 94 employees for a response rate of 97 percent. It was possible to match 90 employee surveys with a manager rating of COBSBs. Thus, the matched response rate was 93 percent. More than 70 percent of the sample held the title of assistant vice president, vice president, or senior vice president, despite their extensive customer interaction. Approximately 75 percent of the sample was male. The median age of service providers was 36. The majority of the sample (88%) reported having a bachelor's degree or higher; 44 percent of the sample had a postgraduate degree. The median organizational tenure was 16 months, considerably lower than the retail banking sample; however, true to their professional orientation, the sample reported having worked in the business lending industry for almost 8 years on average (median = 93 months). All employees were full-time and salaried. In addition, the originators received loan commissions. Nonresponse was not considered a potential threat given the high response rate.

### Measures

**Role stressors.** In both samples, role conflict and role ambiguity were measured with three items each taken from the measures developed by Rizzo et al. (1970). These

measures have been shown to possess good psychometric properties (cf. Kelloway and Barling 1990), and they have been used commonly in marketing studies (cf. Singh et al. 1994). In the business credit sample, one role conflict item had a low item-to-total correlation ( $< .50$ ) and was considered for removal. It was retained for further analyses, however, so that each construct had at least three reflective indicators. Analyses without this item did not alter the results.

**Job satisfaction.** In both samples, job satisfaction was measured using the Minnesota Satisfaction Questionnaire (MSQ) (Weiss, Davis, England, and Lofquist 1967), which consists of 20 positively worded items, measuring intrinsic (11 items), extrinsic (7 items), and global (2 items) satisfaction. This measure has been shown to possess good psychometric properties (Weiss et al. 1967), correlates well with other measures of job satisfaction (Gillet and Schwab 1975), and has been widely used in prior role stressor (cf. Jackson and Schuler 1985) and prosocial behavior studies (cf. MacKenzie et al. 1998). The job satisfaction items were measured on a 5-point scale from 1 (*very dissatisfied*) to 5 (*very satisfied*). Job satisfaction was represented in our structural model analyses as a reflection of three sum scale indicators (intrinsic, extrinsic, and global items).

**Organizational commitment.** In both samples, we measured organizational commitment using four items from the scale developed by Mowday, Steers, and Porter (1979). The complete Mowday et al. (1979) scale measures both affective and behavioral aspects of commitment to the organization. This scale (and modifications of it) has been used frequently in prior marketing research (cf. Singh et al. 1994) and has been shown to possess good psychometric properties (Mathieu and Zajac 1990). The four items used in this research were selected to tap only the *psychological* attachment of the employee to the organization (i.e., affective commitment) because this is thought to be the part of organizational commitment that drives prosocial organizational behaviors (cf. Organ and Ryan 1995). The commitment items were measured on a 7-point scale from 1 (*strongly disagree*) to 7 (*strongly agree*).

**COBSBs.** We developed nine items to measure the three dimensions of COBSBs on the basis of (a) the conceptual definitions of each COBSB dimension previously introduced, (b) indicators of a similar nature from prior prosocial behavior and service quality measures (cf. MacKenzie et al. 1998; Parasuraman et al. 1988; Van Dyne et al. 1994), and (c) interviews with employees of the co-operating organizations to identify relevant context-based behaviors for each COBSB dimension. In focus group interviews in the retail banking context, branch managers were encouraged to identify FSE behaviors either during interactions with customers or behind-the-scenes “that

contribute to the ability of XYZ to provide exceptional service.” In the business credit context, one-on-one depth interviews were conducted with more than 15 FSEs across organizational positions to understand individual positions, the types of customer contact involved with each position, and the behaviors of FSEs “that contribute to exceptional service.” The COBSB items are provided in Table 1. The items were measured on a 7-point scale from 1 (*not at all characteristic of employee/me*) to 7 (*extremely characteristic of employee/me*).

The COBSB measures were administered to both the FSEs and their managers in each sample, and the average of these responses on an item-by-item basis was used to represent the three COBSB constructs in the hypothesis tests. Analytically, the averaging process avoids confounds in measure and trait variance, and it is often used to combine responses in organizational research because “using multiple informants and aggregating their responses into a single composite score helps minimize error” (Van Bruggen, Lilien, and Kacker 2002, p. 470). Although convergence in ratings across sources is often only modest due to differences in rating source perspective, this does not preclude averaging the different ratings because they are essentially treated as formative indicators in which high levels of convergence may not be expected (cf. Diamantopoulos and Winklhofer 2001).<sup>6</sup> Conceptually, this approach is preferred to modeling the rating sources separately because it places focus on the construct of interest rather than the rating source. Furthermore, as formative indicators, omitting either rating source from the construct measure would change the nature of the underlying construct being investigated (cf. Diamantopoulos and Winklhofer 2001). In fact, the lower the convergence among rating sources, the more important it is that sources be averaged to measure the construct of interest, unless there are grounds to conclude that one source is more accurate than the other. This is not possible in this case, however, because both managers and FSEs come with biases and limits to their knowledge concerning the relative degree of individual FSE performance of COBSBs.

## RESULTS

### Measurement Model Analyses

Prior to investigating how structural paths of interest compare across groups, we ran two types of confirmatory measurement analyses to test for configural and full or partial metric invariance across our two service employee samples (cf. Steenkamp and Baumgartner 1998).

**Configural measurement invariance.** Configural invariance is supported across study contexts if evidence is provided by confirmatory factor analyses in both contexts

**TABLE 1**  
Means, Standard Deviations, and Standardized Loadings for the Measures

| Measures and Items                                                                | Retail Banking |      |                  | Business Credit |      |                  |
|-----------------------------------------------------------------------------------|----------------|------|------------------|-----------------|------|------------------|
|                                                                                   | M              | SD   | STD              | M               | SD   | STD              |
| Role conflict                                                                     |                |      |                  |                 |      |                  |
| I receive assignments without adequate resources and materials to complete them.  | 3.60           | 1.72 | .69 <sup>a</sup> | 3.20            | 1.68 | .55 <sup>a</sup> |
| I receive incompatible requests from two or more people.                          | 3.26           | 1.65 | .59 <sup>b</sup> | 2.97            | 1.62 | .87 <sup>b</sup> |
| I have to buck a rule or policy in order to carry out some assignments.           | 3.34           | 1.76 | .55              | 3.16            | 1.75 | .38              |
| Role ambiguity                                                                    |                |      |                  |                 |      |                  |
| I have clear, planned goals and objectives for my job. (R)                        | 5.18           | 1.53 | .61 <sup>a</sup> | 4.84            | 1.60 | .74 <sup>a</sup> |
| I know what my responsibilities are. (R)                                          | 6.03           | 1.17 | .87 <sup>b</sup> | 5.78            | 1.17 | .85 <sup>b</sup> |
| I know exactly what is expected of me. (R)                                        | 5.83           | 1.34 | .95              | 5.43            | 1.41 | .97              |
| Job satisfaction                                                                  |                |      |                  |                 |      |                  |
| Intrinsic satisfaction <sup>c</sup>                                               | 42.2           | 6.38 | .86 <sup>a</sup> | 40.9            | 6.19 | .77 <sup>a</sup> |
| Extrinsic satisfaction <sup>d</sup>                                               | 24.4           | 5.21 | .81 <sup>b</sup> | 24.2            | 5.43 | .88 <sup>b</sup> |
| Global satisfaction <sup>e</sup>                                                  | 7.87           | 1.68 | .61              | 7.79            | 1.49 | .66              |
| Organizational commitment                                                         |                |      |                  |                 |      |                  |
| I find that my values and XYZ's values are very similar.                          | 4.55           | 1.59 | .75 <sup>a</sup> | 5.43            | 1.12 | .63 <sup>a</sup> |
| XYZ really inspires the best in me in the way of job performance.                 | 4.62           | 1.67 | .85 <sup>b</sup> | 4.96            | 1.48 | .87 <sup>b</sup> |
| I really care about the fate of XYZ.                                              | 5.67           | 1.38 | .73 <sup>b</sup> | 6.39            | 0.73 | .56 <sup>b</sup> |
| For me, XYZ is the best of all possible organizations for which to work.          | 4.31           | 1.72 | .86              | 4.60            | 1.46 | .83              |
| External representation                                                           |                |      |                  |                 |      |                  |
| Tells outsiders this is a great place to work.                                    | 4.53           | 1.19 | .86 <sup>a</sup> | 5.09            | 1.05 | .81 <sup>a</sup> |
| Generates favorable goodwill for XYZ.                                             | 5.25           | 1.08 | .88              | 5.83            | 0.71 | .76              |
| Says good things about XYZ to others. <sup>f</sup>                                | 5.12           | 1.12 | .90 <sup>b</sup> | 6.22            | 0.88 | .43 <sup>b</sup> |
| Internal influence                                                                |                |      |                  |                 |      |                  |
| Makes constructive suggestions for service improvement.                           | 5.21           | 1.05 | .86 <sup>a</sup> | 5.18            | 0.97 | .66 <sup>a</sup> |
| Shares creative solutions to customer problems with other team members.           | 5.19           | 1.09 | .93 <sup>b</sup> | 5.46            | 0.81 | .76 <sup>b</sup> |
| Encourages coworkers to contribute ideas and suggestions for service improvement. | 4.72           | 1.17 | .84              | 4.83            | 0.99 | .65              |
| Service delivery                                                                  |                |      |                  |                 |      |                  |
| Follows up in a timely manner to customer requests and problems.                  | 6.00           | 0.71 | .81 <sup>a</sup> | 6.12            | 0.61 | .80 <sup>a</sup> |
| Regardless of circumstances, exceptionally courteous and respectful to customers. | 6.19           | 0.71 | .66              | 6.42            | 0.55 | .70              |
| Follows through in a conscientious manner on promises to customers.               | 6.08           | 0.67 | .90 <sup>b</sup> | 6.19            | 0.54 | .73 <sup>b</sup> |

NOTE: (R) indicates a reverse-coded item. All constructs except job satisfaction were measured on a 7-point scale, where 7 represents greater agreement with the statement. Job satisfaction items were measured on a 5-point scale, where 5 represents greater satisfaction. Customer-oriented boundary-spanning behavior (COBSB) items were slightly reworded for employee self-reports. STD refers to standardized loading.

a. Unstandardized item loading is set to 1 across samples to define the scale of the latent construct in the multigroup measurement and structural model analyses.

b. Unstandardized item loading not constrained across samples in the multi-group measurement and structural model analyses with partial metric invariance.

c. Eleven-item summed subscale from Minnesota Satisfaction Questionnaire.

d. Seven-item summed subscale from Minnesota Satisfaction Questionnaire.

e. Two-item summed subscale from Minnesota Satisfaction Questionnaire.

f. This item was reverse coded for the business credit study. It read "Sometimes criticizes XYZ to outsiders." The descriptive statistics are reported for the reverse-coded item.

All measurement loading estimates are significant at  $p < .001$ . Retail banking:  $\chi^2(df = 188) = 330.92, p < .001$ ; CFI = .95; TLI = .94. Business credit:  $\chi^2(df = 188) = 269.56, p < .001$ ; CFI = .91; TLI = .89.

for (a) the overall fit of the specified model; (b) the significance of salient, nonzero factor loadings; and (c) the discriminant validity of the constructs under investigation. In each sample separately, therefore, a seven-construct confirmatory factor model was estimated using the CALIS procedure of SAS 8.2 (SAS Institute 1989) to assess the measurement properties of the constructs. Evidence of configural invariance is provided by the confirmatory analysis results shown in Tables 1 and 2.

In the retail banking sample, the results indicate that the overall confirmatory factor model fit the data well. The fit indices for the measurement model were  $\chi^2(188, N = 220) =$

330.92,  $p < .001$ ; comparative fit index (CFI) = .95; Tucker-Lewis index (TLI) = .94; root mean square error of approximation (RMSEA) = .06;  $\chi^2/df$  ratio = 1.76. Similarly, in the business credit sample, the overall confirmatory factor model fit the data well. The fit indices for the measurement model were  $\chi^2(188, N = 90) = 269.56, p < .001$ ; CFI = .91; TLI = .89; RMSEA = .07;  $\chi^2/df$  ratio = 1.43. Construct reliability estimates provided in Table 2 also provide evidence of good levels of reliability ( $> .70$ ) for all measures except role conflict. Finally, discriminant validity was assessed by comparison of the squared correlation between two latent constructs with their average

**TABLE 2**  
**Latent Construct Intercorrelations and Reliability Estimates**

| Construct                    | Retail Banking |     |      |       | Business Credit |     |      |       | Correlations     |                   |                     |                              |                            |                       |                     |
|------------------------------|----------------|-----|------|-------|-----------------|-----|------|-------|------------------|-------------------|---------------------|------------------------------|----------------------------|-----------------------|---------------------|
|                              | CR             | AVE | M    | SD    | CR              | AVE | M    | SD    | Role<br>Conflict | Role<br>Ambiguity | Job<br>Satisfaction | Organizational<br>Commitment | External<br>Representation | Internal<br>Influence | Service<br>Delivery |
| Role stressors               |                |     |      |       |                 |     |      |       |                  |                   |                     |                              |                            |                       |                     |
| Role conflict                | .64            | .38 | 10.2 | 3.93  | .63             | .40 | 9.32 | 3.74  | —                | .51               | -.61                | -.57                         | -.34                       | -.07*                 | -.16*               |
| Role ambiguity               | .84            | .68 | 6.96 | 3.51  | .94             | .74 | 7.94 | 3.77  | .34              | —                 | -.53                | -.36                         | -.25                       | -.13*                 | -.11*               |
| Job attitudes                |                |     |      |       |                 |     |      |       |                  |                   |                     |                              |                            |                       |                     |
| Job satisfaction             | .84            | .59 | 74.4 | 11.71 | .82             | .60 | 72.9 | 11.59 | -.53             | -.51              | —                   | .85                          | .68                        | .27                   | .34                 |
| Organizational<br>commitment | .88            | .64 | 19.2 | 5.42  | .85             | .54 | 21.4 | 3.93  | -.37             | -.41              | .70                 | —                            | .69                        | .32                   | .25                 |
| COBSBs                       |                |     |      |       |                 |     |      |       |                  |                   |                     |                              |                            |                       |                     |
| External<br>representation   | .91            | .77 | 14.9 | 3.12  | .72             | .47 | 17.1 | 2.10  | -.25             | -.28              | .71                 | .66                          | —                          | .59                   | .69                 |
| Internal influence           | .91            | .77 | 15.1 | 3.05  | .72             | .56 | 15.5 | 2.23  | .06*             | -.05*             | .37                 | .33                          | .61                        | —                     | .57                 |
| Service delivery             | .83            | .63 | 18.3 | 1.81  | .79             | .47 | 18.7 | 1.43  | .05*             | -.16              | .33                 | .21                          | .48                        | .60                   | —                   |

NOTE: Correlations below the diagonal are for the retail banking sample ( $n = 220$ ). Correlations above the diagonal are for the business credit sample ( $n = 90$ ). CR = composite construct reliability; AVE = average variance extracted; COBSB = customer-oriented boundary-spanning behavior.

\* Correlation is *not* significant at  $p < .05$  (two-tailed  $t$ -test). All other correlations are significant ( $p < .05$ ).

variance extracted (AVE) estimates (Fornell and Larcker 1981). Using the Fornell and Larcker (1981) test, discriminant validity was supported for all possible pairs of constructs in the retail banking sample and all but two pairs of constructs in the business credit sample. Organizational commitment and job satisfaction had a correlation of .85, and external representation and service delivery had a correlation of .69 and only passed less stringent tests of discriminant validity in the business credit sample.<sup>7</sup>

**Metric invariance.** We used Amos 4.0 for our simultaneous multigroup measurement and structural analyses (Arbuckle and Wothke 1999). Metric invariance can be tested by simultaneously estimating parallel confirmatory measurement models across samples and statistically comparing (a) the fit of a model in which the measurement loadings that are not set to be equal to 1 to define the scale of the latent constructs are constrained to be the same across samples with (b) the fit of a model in which these same loadings are unconstrained across samples.<sup>8</sup>

The chi-square fit statistic is 653.74 with 391 degrees of freedom for the constrained two-sample measurement model and 601.12 with 376 degrees of freedom for the unconstrained model. The difference between these two measurement models is significant,  $\chi^2_{\text{diff}} = 52.62$  with 15  $df$ ,  $p < .01$ , indicating that the factor structure is not invariant across the two contexts. Although desirable, full metric invariance is rare and, as such, Steenkamp and Baumgartner (1998) advocated partial metric invariance as acceptable for comparing structural paths across contexts. Specifically, they contended that meaningful comparisons of structural paths can be made “provided that at least one item (other than the one fixed at unity to define the scale of each latent construct) was metrically invariant” (p. 81). Thus, we relaxed the invariance constraints

such that each latent construct only had one loading fixed across contexts in addition to the item loading set to be equal to 1 to define the scale of the latent construct. The chi-square fit statistic is 611.10 with 383 degrees of freedom for this partial constraint measurement model, and the difference between the fit of this model and the fit of the unconstrained model is not significant,  $\chi^2_{\text{diff}} = 9.98$  with 7  $df$ ,  $p > .05$ , providing support for partial metric invariance.<sup>9</sup>

## Multigroup Structural Model Analyses

**Theoretical model estimation.** Initially, in keeping with Singh (2000), a fully restricted theoretical model was tested using Amos 4.0 in which all estimated structural path coefficients were constrained to be invariant across the retail banking and business credit samples. The theoretical multigroup structural model with partial metric invariance fits the data well. The fit indices for this model were  $\chi^2(406, N = 220, N = 90) = 641.28$ ,  $p < .001$ ; CFI = .94; TLI = .93; RMSEA = .04;  $\chi^2/df$  ratio = 1.58. All paths of the theoretical model were significant at  $p < .05$ , with the exception of the paths from organizational commitment to service delivery behaviors ( $p > .10$ ) and from role ambiguity to internal influence behaviors ( $p > .10$ ). Because subsequent tests indicated that these two paths did not differ significantly across samples, they were dropped from the model. This revised model fits the data equally well,  $\chi^2(408, N = 220, N = 90) = 642.58$ ,  $p < .001$ ; CFI = .94; TLI = .93; RMSEA = .04;  $\chi^2/df$  ratio = 1.58. All of the structural paths of this model are significant at  $p < .01$ .

**Structural path invariance tests.** We next tested the equality of structural coefficients across samples. The overall fit for the revised structural model provided a base-

**TABLE 3**  
**Final Multigroup Structural Equation Model Results With Partial Metric Invariance Constraints<sup>a</sup>**

| Structural Path              | Retail Banking                        |                        |       | Business Credit                       |                        |       |
|------------------------------|---------------------------------------|------------------------|-------|---------------------------------------|------------------------|-------|
|                              | Unstandardized Estimates <sup>b</sup> | Standardized Estimates |       | Unstandardized Estimates <sup>b</sup> | Standardized Estimates |       |
| OC → External representation | 0.343                                 | (4.66)                 | .372  | 0.343                                 | (4.66)                 | .413  |
| OC → Internal influence      | 0.168                                 | (2.33)                 | .213  | 0.168                                 | (2.33)                 | .226  |
| OC → Service delivery        | —                                     | —                      | —     | —                                     | —                      | —     |
| JS → External representation | 0.079                                 | (5.02)                 | .420  | 0.044                                 | (2.50)                 | .310  |
| JS → Internal influence      | 0.042                                 | (2.51)                 | .262  | 0.042                                 | (2.51)                 | .335  |
| JS → Service delivery        | 0.042                                 | (4.78)                 | .400  | 0.042                                 | (4.78)                 | .425  |
| JS → OC                      | 0.143                                 | (10.46)                | .700  | 0.143                                 | (10.46)                | .842  |
| RC → JS                      | -2.036                                | (-4.99)                | -.428 | -2.036                                | (-4.99)                | -.390 |
| RA → JS                      | -1.782                                | (-5.12)                | -.326 | -1.782                                | (-5.14)                | -.373 |
| RC → Internal influence      | 0.160                                 | (2.52)                 | .209  | 0.160                                 | (2.52)                 | .232  |
| RA → Internal influence      | —                                     | —                      | —     | —                                     | —                      | —     |
| RC → Service delivery        | 0.078                                 | (1.77)                 | .157  | 0.078                                 | (1.77)                 | .146  |

NOTE: RC = role conflict; RA = role ambiguity; JS = job satisfaction; OC = organizational commitment.

a. Fit indices are as follows:  $\chi^2(406, N = 220, N = 90) = 634.93$ ; Comparative Fit Index (CFI) = .94; Tucker-Lewis Index (TLI) = .93.

b. The structural estimates reported are from a multigroup analysis in which the structural model was estimated simultaneously with partial metric invariance constraints for the retail banking and business credit samples using Amos 4.0 (see also Note 9). *t*-values are shown in parentheses. Based on one-tailed *t*-tests: *t*-values/  $\geq 2.33$  are significant at .01 and *t*-values/  $\geq 1.65$  are significant at .05 for the retail banking sample. *t*-values/  $\geq 2.38$  are significant at .01 and *t*-values/  $\geq 1.67$  are significant at .05 for the business credit sample. Coefficients that differ significantly ( $p < .05$ ) across the two samples are in italics.

line for nested model comparisons of more and less restricted structural models in which the invariance constraints placed on structural path coefficients were released sequentially. Comparison of the fit of the revised structural model with the fit of each of the nine other nested multigroup models in which the nine significant constrained paths of the revised structural model were released one at a time revealed only one significant difference in fit with a one degree of freedom difference between models. These model comparisons revealed that the path from job satisfaction to external representation was stronger in the retail banking context ( $\beta_{\text{unstd}} = .08, p < .01$ ) than it was in the business credit context ( $\beta_{\text{unstd}} = .04, p < .01$ ),  $\chi^2_{\text{diff}} = 4.41, df = 1; p < .05$ , as shown in Table 3. However, this path is still significant in both samples. Comparisons of the fit between the revised structural model and each of the six models in which a path constrained to be zero in Figure 1 was added one at a time revealed no significant differences in fit with a one degree of freedom difference between models. However, these comparisons did reveal that the path from role conflict to service delivery behaviors was significant and should be added to the revised multigroup model ( $\beta_{\text{unstd}} = .08, p < .05$ ), as shown in Table 3, despite a nonsignificant change in the chi-square statistic,  $\chi^2_{\text{diff}} = 3.09, df = 1; p > .05$ . Subsequent comparisons of each of the zero-constrained paths of Figure 1, including the role conflict to service delivery path, revealed that none was significantly different across samples.

*Tests of the hypothesized relationships between constructs.* The final multigroup structural model with partial metric invariance constraints fits the data well and is presented in Table 3. The fit indices for this model were  $\chi^2(406, N = 220, N = 90) = 634.93, p < .001$ ; CFI = .94; TLI = .93; RMSEA = .04;  $\chi^2/df$  ratio = 1.56. All structural paths of the final model were significant at  $p < .05$ , including the path from job satisfaction to external representation in each sample. To test the hypotheses, the indirect effects of the role stress constructs on the COBSBs must be tested. Table 4 shows the standardized and unstandardized estimates corresponding to each of the indirect effects of the role conflict and role ambiguity constructs on the COBSB constructs and a *z*-value testing whether each of these indirect effects is significantly different from zero. As shown, the hypothesized negative indirect effects of the role stress constructs on external representation, internal influence, and service delivery were all significant, thus supporting Hypothesis 1a (retail banking:  $z = -4.66, p < .001$ ; business credit:  $z = -4.33, p < .001$ ), Hypothesis 1b ( $z = -3.52, p < .001$ ), Hypothesis 1c ( $z = -3.38, p < .001$ ), Hypothesis 2a (retail banking:  $z = -4.77, p < .001$ ; business credit:  $z = -4.41, p < .001$ ), Hypothesis 2b ( $z = -3.92, p < .001$ ), and Hypothesis 2c ( $z = -3.68, p < .001$ ).

Hypotheses 3a and 3b involved predictions about the relative magnitudes of the indirect effects of the role stress constructs on specific forms of COBSBs. It was expected that the indirect negative effects of the role stressors on

**TABLE 4**  
**Indirect Effects of Role Conflict and Role Ambiguity on COBSBs Based on Final Multigroup Structural Equation Model Results With Partial Metric Invariance Constraints**

| <i>Hypothesized Indirect Structural Relationships</i>                 | <i>Retail Banking</i>          |                              |                | <i>Business Credit</i>         |                              |                | <i>Conclusions</i>                    |
|-----------------------------------------------------------------------|--------------------------------|------------------------------|----------------|--------------------------------|------------------------------|----------------|---------------------------------------|
|                                                                       | <i>Unstandardized Estimate</i> | <i>Standardized Estimate</i> | <i>z-Value</i> | <i>Unstandardized Estimate</i> | <i>Standardized Estimate</i> | <i>z-Value</i> |                                       |
| Indirect negative effect of role conflict on external representation  | -.261                          | -.291                        | -4.66          | -.189                          | -.245                        | -4.33          | Hypothesis 1a supported, both samples |
| Indirect negative effect of role conflict on internal influence       | -.135                          | -.176                        | -3.52          | -.135                          | -.196                        | -3.52          | Hypothesis 1b supported, both samples |
| Indirect negative effect of role conflict on service delivery         | -.085                          | -.171                        | -3.38          | -.085                          | -.158                        | -3.38          | Hypothesis 1c supported, both samples |
| Indirect negative effect of role ambiguity on external representation | -.229                          | -.222                        | -4.77          | -.165                          | -.256                        | -4.41          | Hypothesis 2a supported, both samples |
| Indirect negative effect of role ambiguity on internal influence      | -.118                          | -.134                        | -3.92          | -.118                          | -.205                        | -3.92          | Hypothesis 2b supported, both samples |
| Indirect negative effect of role ambiguity on service delivery        | -.075                          | -.130                        | -3.68          | -.075                          | -.166                        | -3.68          | Hypothesis 2c supported, both samples |

external representation (Hypothesis 3a) and internal influence (Hypothesis 3b) would be stronger than the indirect negative effects of the role stressors on service delivery. As shown in Table 4, the relative magnitudes of the indirect structural coefficients of each of the role stress constructs on the COBSB dimensions is consistent with the hypotheses because the indirect negative effects of role conflict on external representation (retail banking:  $\beta_{\text{unstd}} = -.26, p < .001$ ; business credit:  $\beta_{\text{unstd}} = -.25, p < .001$ ) and internal influence ( $\beta_{\text{unstd}} = -.14, p < .001$ ) were both significant and larger than the indirect effect of role conflict on service delivery ( $\beta_{\text{unstd}} = -.09, p < .001$ ). Similarly, the indirect negative effects of role ambiguity on external representation (retail banking:  $\beta_{\text{unstd}} = -.23, p < .001$ ; business credit:  $\beta_{\text{unstd}} = -.26, p < .001$ ) and internal influence ( $\beta_{\text{unstd}} = -.12, p < .001$ ) were both significant and larger than the indirect effect of role ambiguity on service delivery ( $\beta_{\text{unstd}} = -.08, p < .001$ ).

To provide a statistical test of Hypotheses 3a and 3b, we conducted nested model comparisons with certain imposed equality constraints on the direct paths from job satisfaction and organizational commitment to the COBSB dimensions being compared. Specifically, to test Hypothesis 3a (Hypothesis 3b), we estimated a model (a) in which the direct paths from job satisfaction to external representation (internal influence) and from job satisfaction to service delivery were constrained to be equal and (b) in which the direct paths from organizational commitment to external representation (internal influence) and from organizational commitment to service delivery were constrained to be equal, and we compared the chi-square fit of this model with the fit of a baseline structural model in which these constraints were not imposed. Such equality constraints on the direct paths from job satisfaction and organizational commitment to COBSBs provide a

comparison of differences in the indirect effects of the role stressors on different COBSB dimensions because the indirect paths from the role stressors to the distinct COBSB dimensions only differ in these direct paths. For Hypothesis 3a, the equality constraint on the path from job satisfaction to external representation was imposed one group at a time because this direct path was previously determined to be significantly different across samples.

The baseline model estimated for the statistical tests of Hypotheses 3a and 3b was the final multigroup structural model of Table 3 with the addition of the path from organizational commitment to service delivery behaviors (constrained to be equal across samples). The path from organizational commitment to service delivery was added back to the model for the tests of Hypotheses 3a and 3b so that there would be parallel paths leading into each distinct COBSB dimension.<sup>10</sup> The difference in fit between this baseline structural model and the constrained model for each sample to test Hypothesis 3a was significant (retail banking:  $\chi^2_{\text{diff}} = 65.13, df = 2; p < .01$ ; business credit:  $\chi^2_{\text{diff}} = 32.77, df = 2; p < .01$ ), providing support for Hypothesis 3a. Similarly, the difference in fit between the baseline structural model and the constrained model to test Hypothesis 3b was significant ( $\chi^2_{\text{diff}} = 11.35, df = 2; p < .01$ ), providing support for Hypothesis 3b. Thus, these analyses provide support for stronger indirect negative effects from the role stressors to external representation and internal influence in comparison to service delivery behaviors.

Hypotheses 4a and 4b involved predictions about positive direct effects of role conflict and role ambiguity on internal influence after accounting for negative indirect effects of the role stress constructs on internal influence behaviors via job satisfaction and/or organizational commitment. As shown in Table 3, the direct path from role conflict to internal influence behaviors is significant and

positive ( $\beta_{\text{unstd}} = .16, p < .01$ ), providing support for Hypothesis 4a. However, the direct path from role ambiguity to internal influence behaviors was not significant, and it did not significantly differ across samples ( $p > .05$ ). Thus, support is not provided for Hypothesis 4b. However, the final structural model also includes a direct positive path from role conflict to service delivery behaviors that was added to the theoretical model ( $\beta_{\text{unstd}} = .08, p < .05$ ).

*Assessment of the effects of common method variance.* Because all of the data were obtained at least partially from the service employees in each study, common method variance provides a potential rival explanation for the observed relationships between the constructs. As noted by Cote and Buckley (1987), common method variance can increase, decrease, or have no effect on observed relationships between latent constructs. To investigate the extent to which this potential source of bias may have affected the strength of the relationships observed, we refit the final multigroup structural model, with the addition of a first-order method factor in each study context. For each sample, this extra factor was defined as loading on all of the indicators of the latent constructs and thus captures any additional systematic method variance common to these measures in each study context (e.g., common rater bias, social desirability, "yea saying," etc.).

Overall, the fit indices,  $\chi^2(362, N = 220, N = 90) = 498.70, p < .001$ ; CFI = .96; TLI = .95; RMSEA = .04;  $\chi^2/df$  ratio = 1.38, for this model suggest that it fit the data significantly better than the final multigroup model reported in Table 3, due to the fact that the method factor was significantly related to 16 of the self-report measures in the retail banking context and 8 of the self-report measures in the business credit context. Perhaps more important, the results indicate that (a) the addition of the method factor did not cause *any* of the hypothesized measurement relationships to become nonsignificant in either sample, and (b) the hypothesized factor loadings were consistently much higher than the method factor loadings in each sample. However, the results did reveal that two structural paths were attenuated to the point of nonsignificance by the presence of the method factor. Specifically, the paths from role conflict to service delivery and from organizational commitment to internal influence become nonsignificant as a result of accounting for common method variance in the final multigroup structural model. All other structural paths remain significant in the direction hypothesized.

## DISCUSSION

### General Discussion of Results

The theoretical model we developed and tested relies on prior models of employee behavioral withdrawal (e.g.,

turnover) as its basis (cf. Netemeyer et al., 1995). Across both of our service contexts, our results demonstrated that role conflict and role ambiguity lead to psychological withdrawal from the job and organization in terms of reduced job satisfaction and organizational commitment, which in turn lead to reduced performance of COBSBs. Our results also provided support for our hypotheses concerning the differential strength of the negative effects of role conflict and role ambiguity on different dimensions of COBSBs. Consistent with the social exchange reasoning of Organ (1988b), the results revealed that both role conflict and role ambiguity had stronger negative relationships through job satisfaction and organizational commitment with external representation and internal influence behaviors than they did with service delivery behaviors. Finally, our results demonstrate that once the effects of common method variance were controlled, the direct positive effects of role conflict on COBSBs were limited to the internal influence behaviors. This finding reinforces the importance of future research considering possible differential predictors of distinct dimensions of COBSBs.

It is also important that we comment specifically on the parallel nature of our results across two very distinct service delivery contexts. Structural invariance tests revealed only one path that was significantly different across the retail banking and business credit contexts, despite many differences between the service contexts and employee samples. Specifically, the positive path relationship between job satisfaction and external representation behaviors was stronger in the retail banking context than in the business credit context. In terms of understanding the withdrawal effects of role stressors, this path difference means that the indirect negative effects of both role conflict and role ambiguity on external representation behaviors through job satisfaction were stronger in the retail banking context. The retail banking employees were more likely to withdraw their advocacy of the organization when faced with increased levels of role conflict and role ambiguity.

### Limitations and Future Research

In considering our results, some limitations of our studies should be kept in mind. First, although we investigated a structural model of relationships among variables, our research design was cross-sectional. We can only conclude that our model is a feasible explanation of the observed relationships in the data. Causality cannot be concluded. Rather, the direction of paths in our model rely on prior empirical results and theoretical arguments. Second, although we investigated our theoretical model using two very distinct service contexts and samples, the generalizability of our findings beyond these contexts relies on additional research.

In addition to addressing these methodological limitations, there are also other research opportunities related to our findings. First, future research should explore the reasons behind the simultaneous positive and negative effects of role conflict on internal influence behaviors. One possibility that should be investigated is that different dimensions of role conflict may have distinct relationships with internal influence behaviors. Singh (2000) has shown that different dimensions of role stress can have different relationships with the attitudinal and behavioral outcomes of boundary-spanning employees. It is possible, for example, that boundary-spanning employees may respond constructively to role conflict introduced by customers but psychologically and behaviorally withdraw from role conflict created by the supervisors or policies of the organization. Another possibility that should be investigated is why some types of employees might respond constructively to role conflict, whereas others might respond with psychological and behavioral withdrawal. Griffeth et al. (1999), for example, have proposed that employee responses to objectionable work experiences will depend on where they fall in a  $2 \times 2$  matrix defined by an employee's level of job involvement and organizational commitment. They postulate that an institutionalized star, an employee with high job involvement and high organizational commitment, is more likely to respond constructively through vocal attempts to change the work situation than other types of employees.

Future research is also recommended to better understand the reasons for the differential strength of the indirect path relationships from the role stressors to distinct dimensions of COBSBs. We have offered a rationale derived from social exchange theory based on the degree to which each form of COBSB is perceived to be extrarole by employees. Social exchange theory has face validity as an explanation of our findings in light of internal company documents and past conceptual thinking on citizenship behaviors. Furthermore, even the means and variance of each dimension of COBSBs hint at the validity of this rationale (i.e., service delivery behaviors have higher means and lower variance across employees as might be expected from more in-role behaviors). Future empirical work can more explicitly test the social exchange rationale by explicitly seeking employee input into the degree to which each behavior is perceived to be extrarole. Finally, additional research is needed concerning the antecedents of COBSBs. In particular, future research may rely on the customer-oriented nature of COBSBs to guide selection of antecedents.

### Managerial Implications

For quite some time, it has been recommended that managers take actions to reduce role conflict and role

ambiguity in order to improve traditional performance measures. Our research extends this work to demonstrate that these same actions can also improve the customer orientation of the organization by removing workplace conditions that serve to decrease or inhibit performance of customer-oriented behaviors. Although our results show that reduced role conflict and role ambiguity will lead to increased performance of customer-oriented behaviors through their impact on job satisfaction and organizational commitment rather than directly, the implications for managers are the same. In combination with our results, past research suggests that customer-oriented behaviors of FSEs can be enhanced by such organizational and managerial practices as offering frequent and specific feedback concerning performance, providing structured leadership guidance, implementing clear policies, and facilitating participation in decision making among frontline employees that lead to reductions in role conflict and role ambiguity (cf. Jackson and Schuler 1985), even if these practices are not specific to customer-oriented behaviors.

Finally, our results also suggest that managers should seek to reduce both role conflict and role ambiguity to minimum levels. Past research has suggested that this might be a misguided approach because of potentially competing negative and positive effects of role conflict and role ambiguity on global performance measures. However, when considering separate facets of FSE customer-oriented behaviors, our results demonstrate that the only positive effect of the two role stressors on COBSBs was the direct path from role conflict to internal influence. While it is desirable to encourage such constructive change efforts, it is also likely that these efforts are geared to changing current conflicting situations with regard to serving customers (cf. Hirschman 1970). Therefore, simply reducing role conflicts in the first place or helping employees to understand how to work around and through them would reduce the need for internal influence behaviors while at the same time reducing the negative withdrawal effects of role conflict in the workplace on each dimension of COBSBs.

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## NOTES

1. Literature on pro-social organizational behaviors, including organizational citizenship research (cf. Bettencourt, Gwinner, and Meuter 2001), has had considerable influence on our conceptualization, measures, model, and hypotheses. For example, a three-dimension conceptualization of organizational citizenship introduced by Van Dyne, Graham, and Dienesch (1994) closely parallels our three-dimension customer-oriented behavior conceptualization.

2. Performance in its broadest sense refers simply to behaviors that are assumed to lead to the attainment of organizational goals. Thus, the "performance" label may be applied broadly to refer to either in-role or extrarole behaviors, as we do in our conceptual discussion.

3. From a statistical perspective, Bollen (1989) stated that the failure to include intervening variables can lead to misleading results when

the direct and indirect effects for a more fully specified model are of opposite signs but similar magnitudes. Then a left-out intervening variable can lead to the absolute value of a total effect considerably less than the absolute values of the direct or indirect effects of which it is comprised. (P. 47)

4. A potential concern here is the absence of burnout in our theoretical model and the possibility this raises for a misspecified model. Although model misspecification cannot be entirely ruled out, substantial prior theoretical and empirical research suggests that the addition of burnout to the model would not have (1) accounted for the relationships between the other constructs in the model, nor would it necessarily have been (2) a preferable mediator for separating the competing positive and negative effects of role stressors on COBSBs.

5. The number of employees included in the retail banking sample is estimated based on the total number of employees filling the two positions within the 106 branches sampled.

6. For interested readers, the correlations between self-ratings and supervisor ratings ranged from .16 to .34 in the retail banking context and from .06 to .22 in the business credit context.

7. First, the 95 percent confidence interval for the correlation between each pair of constructs was not found to include unity ( $SE = .05$  for commitment-satisfaction correlation,  $SE = .09$  for external representation-service delivery correlation). Second, nested measurement models in which the correlation between each latent construct pair was set separately to unity produced significant changes in chi-square model fit (commitment-satisfaction:  $\chi^2_{diff} = 14.18$ ,  $df = 1$ ; external representation-service delivery:  $\chi^2_{diff} = 18.34$ ,  $df = 1$ ).

8. Five customer-oriented boundary-spanning behavior indicators originally included among the items in the retail banking context were not relevant in the business credit context and were, therefore, not included in the multigroup analyses. Construct validity with these additional items remains strong. More important, the statistical significance and direction of the structural paths remain unchanged.

9. All subsequent analyses and statistical comparisons were also investigated using full metric invariance constraints. These additional analyses reveal consistent results with those reported using partial metric invariance constraints.

10. It was important to add this path back to the model for the nested model comparisons of Hypotheses 3a and 3b because concluding that one path is not significantly different from zero whereas another path is significantly different from zero is not the same as concluding that these same two paths significantly differ from one another (the conclusion essentially being tested here).

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